Analyzing Non-Normal Data

AGRON 5130

YOUR NAME HERE

Introduction

The goal of this assignment is to give you hands-on practice with **diagnosing model assumptions** and working with **non-normal data**.

You will work with a grape yield dataset from a vineyard experiment evaluating canopy management strategies.

In this assignment, you will practice:

- visualizing skewed data
- diagnosing violations of model assumptions
- applying a **log transformation**
- fitting a **GLM**
- interpreting treatment effects across models

Submit a **knitted PDF/HTML generated from your R Markdown file** to Canvas.

Background

Grapevine productivity is strongly influenced by canopy structure, which affects light interception, air movement, and ultimately fruit development.

In commercial vineyards, growers often use different **canopy management practices** to balance vegetative growth and fruit production. However, these practices can have variable effects depending on environmental conditions and vine-to-vine variability.

A field trial was conducted to evaluate the effect of different canopy management strategies on grapevine productivity in a mature vineyard.

The treatments included:

- **Control** (no canopy management)

- **Shoot thinning** (removal of excess shoots early in the season)

- **Leaf removal** (removal of basal leaves around the fruiting zone)

- **Shoot thinning + leaf removal**

The experiment was conducted using a **completely randomized design** with 30 replications.

At the end of the growing season, the number of **grape clusters per vine** was recorded for each experimental unit.

Due to natural variability among vines and the biological processes governing fruit set, the response variable is expected to show:

- substantial variability across treatments
- increasing variance with increasing mean
- right-skewed distributions

These characteristics make this dataset a good candidate for exploring **violations of model assumptions** and methods to address them.

We can start by loading the necessary libraries

```
library(ggplot2)
library(emmeans)
```

Question 1

1 point

Load the dataset `"data/grape.csv"` and display the first few rows.

```
grape <- read.csv('./data/grape.csv')
head(grape)
```

```
##   treatment clusters
## 1   Control        1
## 2   Control       21
## 3   Control        2
## 4   Control        3
## 5   Control        2
## 6   Control        1
```

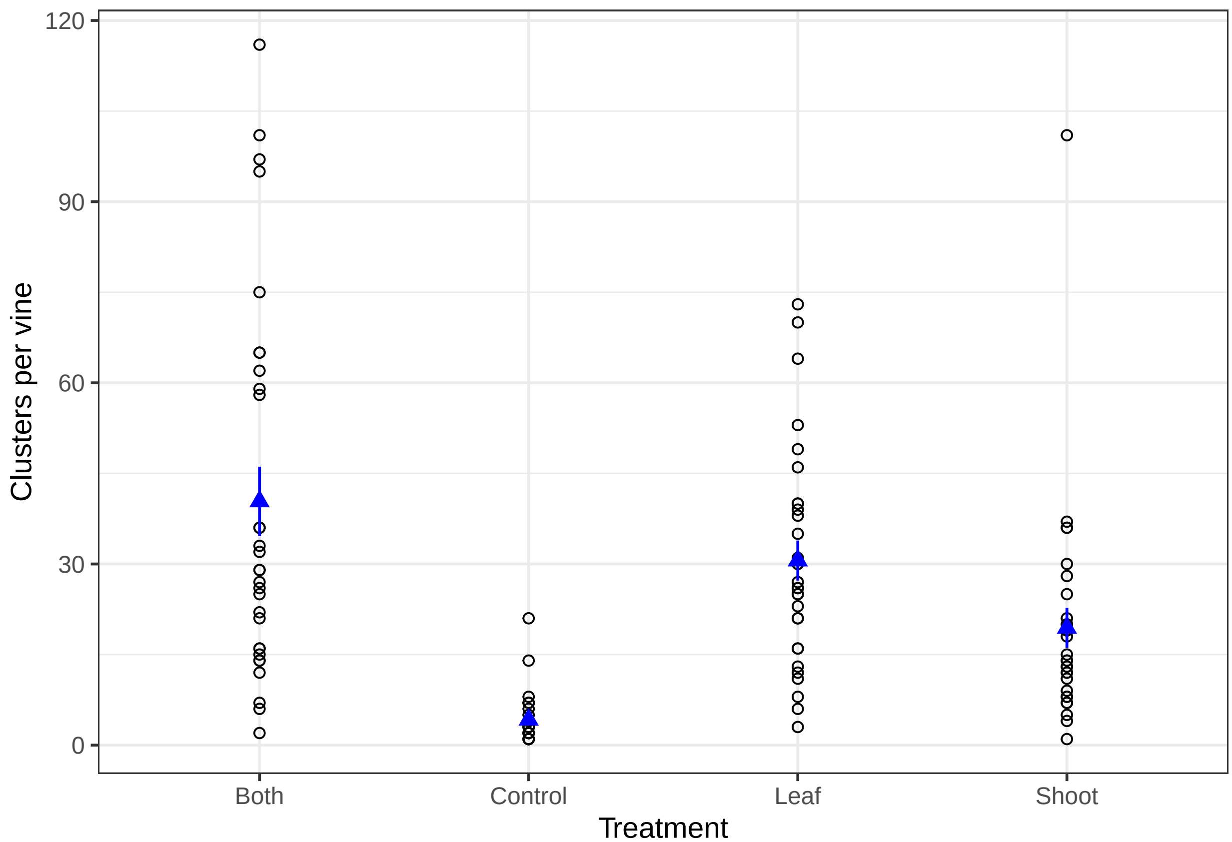
Question 2

2 points

Create a plot showing the relationship between `treatment` and `clusters`.

```
ggplot()+
  geom_point(data = grape,
             aes(x = treatment, y = clusters),
             pch = 1)+
  stat_summary(data = grape,
               fun.data = 'mean_se',
               aes(x = treatment, y = clusters),
               col = 'blue',
```

```
             pch = 17)+
  labs(y = 'Clusters per vine', x = 'Treatment')+
  theme_bw()
```



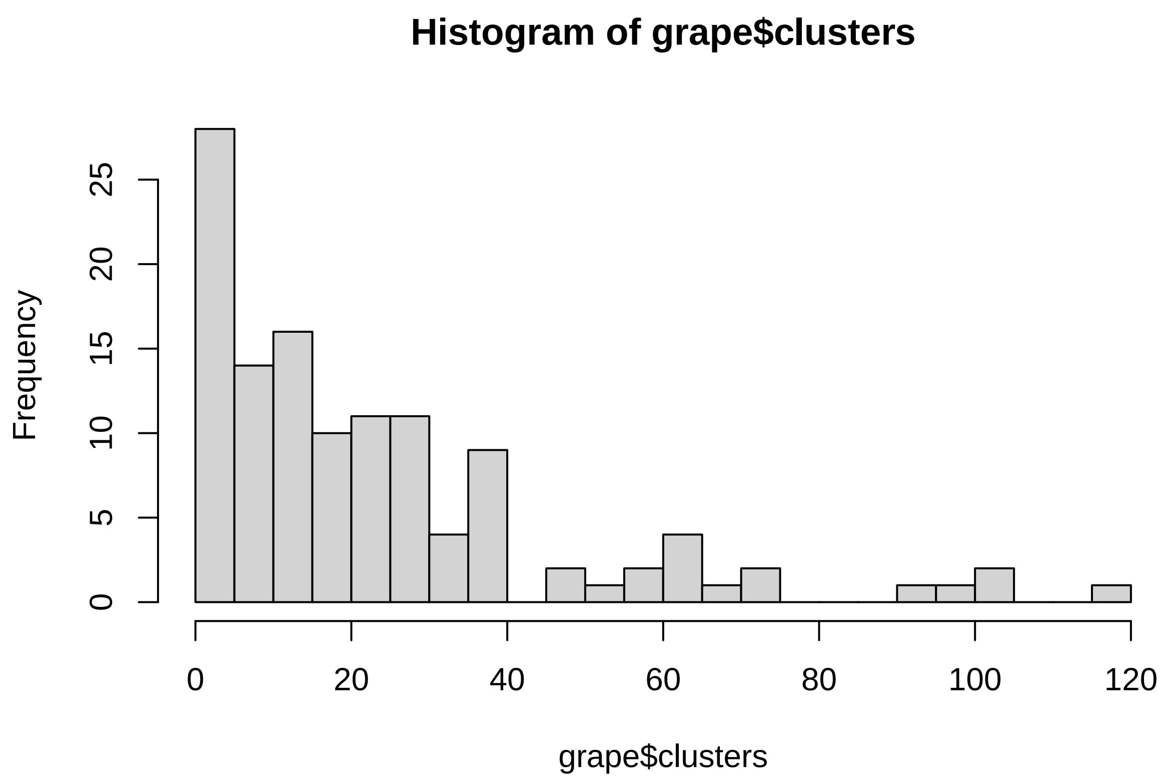
Question 3

1 point

Create a histogram of `clusters`.

Briefly describe the distribution (normal vs skewed).

```
hist(grape$clusters,
     breaks = 20)
```



Question 4

2 points

Fit a linear model and extract the treatment means. Then, comment on the confidence intervals. Do any of the confidence intervals include negative values? Why might this be a problem for this type of data?

```
model1 <- lm(clusters ~ treatment,
             data = grape)
anova(model1)
```

```
## Analysis of Variance Table
##
## Response: clusters
##            Df Sum Sq Mean Sq F value    Pr(>F)
## treatment   3  21710  7236.5  17.377 2.197e-09 ***
## Residuals 116  48308   416.5
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
emmeans(model1, ~ treatment)
```

```
##  treatment emmean   SE  df lower.CL upper.CL
##  Both        40.4 3.73 116    32.99     47.7
```

```
##   Control       4.2 3.73 116    -3.18     11.6
##   Leaf         30.6 3.73 116    23.19     37.9
##   Shoot        19.4 3.73 116    12.02     26.8
##
## Confidence level used: 0.95
```

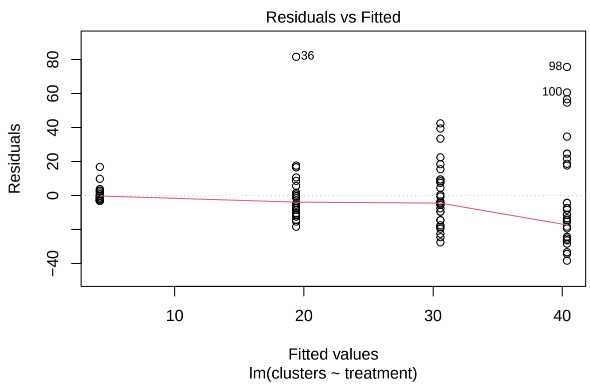
Question 5

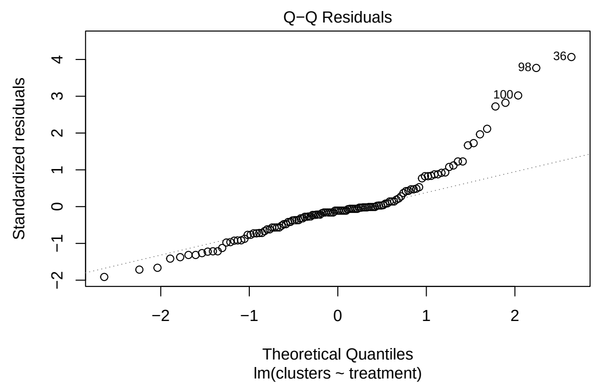
1 points

Create diagnostic plots and comment on:

- normality
- constant variance

```r
plot(model1, which = 1)
plot(model1, which = 2)
```





Question 6

1 points

Fit a model using a **log transformation** of the response.

```r
ml <- lm(log(clusters) ~ treatment,
         data = grape)
anova(ml)
```

```
## Analysis of Variance Table
##
## Response: log(clusters)
##            Df Sum Sq Mean Sq F value    Pr(>F)
## treatment   3 96.147  32.049  46.725 < 2.2e-16 ***
## Residuals 116 79.564   0.686
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

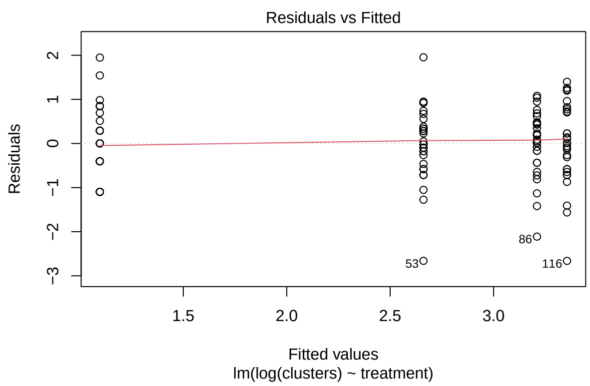
Question 7

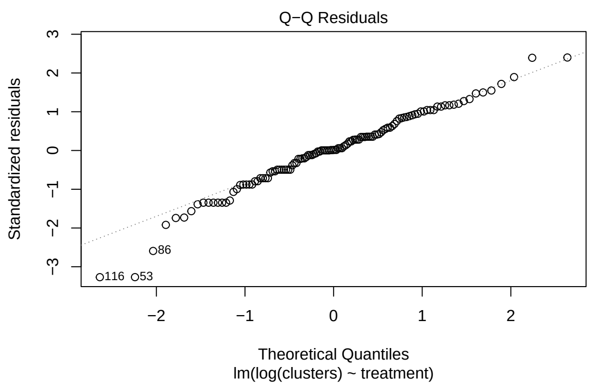
1 point

Create diagnostic plots for the log-transformed model.

Did things improve?

```r
plot(ml, which = 1)
plot(ml, which = 2)
```





Question 8

1 point

Compute estimated marginal means for the log model.

```r
log_means <- emmeans(ml, ~ treatment)
log_means <- data.frame(log_means)
log_means
```

```
##   treatment   emmean        SE  df  lower.CL upper.CL
## 1      Both 3.354933 0.1512062 116 3.0554501 3.654416
## 2   Control 1.096407 0.1512062 116 0.7969244 1.395890
## 3      Leaf 3.209876 0.1512062 116 2.9103935 3.509359
## 4     Shoot 2.661082 0.1512062 116 2.3615992 2.960565
```

Question 9

1 point

Back-transform the means using `exp()`.

```r
exp(log_means$emmean)
```

```
## [1] 28.643684  2.993392 24.776023 14.311767
```

Question 10

1 point

Compute the difference between the treatments `Both` and `Control` presented on the log scale then interpret this difference in the data scale

```
exp(3.354933 - 1.096407)
```

```
## [1] 9.568974
```

```
28.643684 / 2.993392
```

```
## [1] 9.568972
```

Now we will fit a **GLM appropriate for this type of data** using a **Gamma distribution with a log link**. We will use the model `mod_glm` for the next questions.

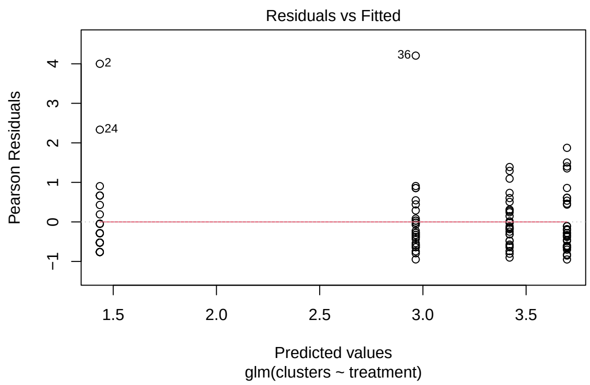
```
mod_glm <- glm(clusters ~ treatment,
               family = Gamma(link = "log"),
               data = grape)
```

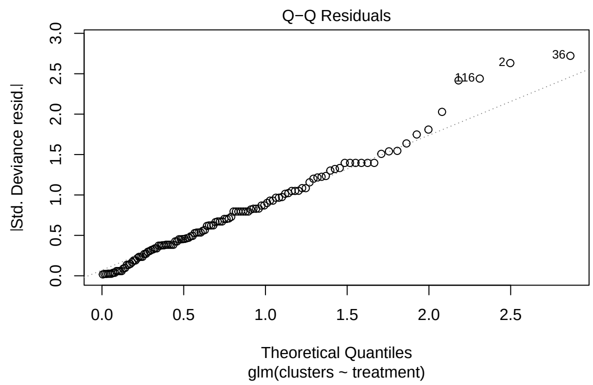
Question 11

1 point

Create diagnostic plots for the GLM model. Is this model a good fit? Are we concerned about violating assumptions?

```
plot(mod_glm, which = 1)
plot(mod_glm, which = 2)
```





Question 12

1 point

Use `emmeans()` to extract treatment means on the **response scale**. Then, comment on the confidence intervals. Do any of the confidence intervals include negative values?

```
emmeans(mod_glm, ~ treatment, type = 'response')
```

```
##  treatment response    SE  df lower.CL upper.CL
##  Both          40.4 6.230 116    29.74     54.8
##  Control        4.2 0.648 116     3.09      5.7
##  Leaf          30.6 4.720 116    22.52     41.5
##  Shoot         19.4 2.990 116    14.29     26.3
##
## Confidence level used: 0.95
## Intervals are back-transformed from the log scale
```

Question 13

1 point

Briefly compare the GLM results to the log-transformed model (are the treatment effects similar?). Discuss how these two approaches are an improvement uppon the first approach in which we ignore the non-normality of the residuals.